

Volatility Modeling

The goal of this project is to combine econometric techniques learned in this course with modern machine-learning tools to develop a trading strategy based on volatility in the energy sector. Specifically, we aim to investigate whether incorporating weather anomalies can improve forecasts of energy-sector volatility, and whether these improved forecasts translate into superior trading performance.

To structure the analysis, the project naturally divides into three components:

1. Weather Anomaly Detection
2. Volatility Forecasting
3. Trading Strategy

Below, we focus on the volatility forecasting component, as it serves as the core link between weather-driven signals and actionable market decisions. This section of the project is responsible for taking in some exogenous features for weather anomalies and then outputting a predicted volatility for the next period

1 Volatility Measure

To begin, we specify how volatility in the energy sector is measured. This process consists of two steps. First, we isolate energy-sector returns that are orthogonal to movements in the overall market. Second, we define an appropriate measure of volatility for these residual returns.

Let $\{P_t^{SPY}\}$ denote the price series of the S&P 500 ETF (SPY), and let $\{P_t^{XLE}\}$ denote the price series of the energy-sector ETF (XLE).

We compute log returns for each series to ensure symmetry and time additivity:

$$r_t = \log\left(\frac{P_t}{P_{t-1}}\right).$$

Then extracting the portion of energy-sector returns not explained by broad market movements, we estimate the following regression:

$$r_t^{XLE} = \beta_0 + \beta_1 r_t^{SPY} + \varepsilon_t.$$

The regression residuals ε_t represent the idiosyncratic component of XLE returns. We denote this series by φ_t :

$$\varphi_t = r_t^{XLE} - \hat{r}_t^{XLE}.$$

We now define our measure of volatility. In the volatility forecasting literature, this task is challenging because true latent volatility is unobserved. A common and effective proxy is realized volatility, constructed from high-frequency return data. [4], [3]

Let $\{\varphi_{t,i}\}_{i=1}^n$ denote the intraday idiosyncratic returns on day t , computed over non-overlapping two-hour intervals, where n is the total number of such intervals within a trading day. We define daily realized volatility as the square root of the scaled sum of squared intraday returns:

$$RV_t = \sqrt{\sum_{i=1}^n \varphi_{t,i}^2}.$$

2 Evaluating Performance

To assess the predictive performance of our volatility models, we adopt an expanding-window forecasting scheme combined with a one-step-ahead prediction structure [1]. Let $\{RV_t\}_{t=1}^T$ denote the realized volatility series, where RV_t is the target variable of interest.

We partition the sample into three components: a training set, a validation set, and a final test set. The training set is used to estimate model parameters, the validation set is employed for model comparison, hyperparameter tuning, and selection of relevant exogenous features, and the test set is reserved strictly for out-of-sample performance evaluation.

Let $\mathcal{F}_t$ denote the information set available at time t , consisting of all past returns and lagged exogenous variables. Specifically,

$$\mathcal{F}_t = \sigma(\{r_s\}_{s \leq t} \cup \{\mathbf{x}_{s-1}\}_{s \leq t}),$$

so that returns at time t depend only on exogenous covariates observed at time $t - 1$. All forecasts are generated conditionally on $\mathcal{F}_t$, ensuring that model estimation, tuning, and evaluation respect the information constraints faced by an investor in real time.

Expanding Window Procedure. Suppose the total sample has T observations, and the initial estimation window contains T_0 observations (e.g., $T_0 = 80$ if $T = 100$). The expanding-window method proceeds as follows:

1. Estimate the model using observations 1 through T_0 .
2. Generate a one-step-ahead forecast given the information $\hat{\sigma}_{t+1|t} = \sqrt{\mathbb{E}[\varphi_{t+1}^2 | \mathcal{F}_t]}$.
3. Expand the estimation window to include the new observation, i.e., re-estimate the model using observations 1 through $T_0 + 1$.
4. Produce the next one-step-ahead forecast $\hat{\sigma}_{t+1|t}$.

This process continues until the end of the sample, yielding a sequence of out-of-sample one-step-ahead forecasts $\{\hat{\sigma}_t\}_{t=T_0+1}^T$.

Forecast Evaluation via QLIKE. To evaluate forecast accuracy, we employ the QLIKE loss function, a standard metric in the volatility-forecasting literature between the forecasted and actual volatility measures.

$$\text{QLIKE}(RV_t, \hat{\sigma}_t) = \frac{RV_t}{\hat{\sigma}_t} - \log\left(\frac{RV_t}{\hat{\sigma}_t}\right) - 1.$$

Given the sequence of forecasts, the overall out-of-sample loss is computed as

$$\text{QLIKE}_{\text{OOS}} = \frac{1}{T - T_0} \sum_{t=T_0+1}^T \text{QLIKE}(RV_t, \hat{\sigma}_t).$$

This metric is robust to measurement error in realized volatility and is widely used in comparing volatility models. The model with the lowest average QLIKE loss is considered the best-performing specification.

3 AR-X GARCH Models

Financial return series are characterized by time-varying volatility, with periods of high variability followed by periods of relative calm. To capture this feature, we model returns using a conditionally heteroskedastic framework [2] in which the variance of returns evolves over time as a function of past information.

Let $\{r_t\}$ denote a return series. Returns are assumed to follow

$$r_t = \mu_t + \varepsilon_t, \quad \varepsilon_t = \sigma_t z_t,$$

where μ_t is the conditional mean, σ_t^2 is the conditional variance, and z_t is an innovation satisfying $\mathbb{E}[z_t] = 0$ and $\mathbb{V}[z_t] = 1$.

The conditional variance is modeled using a generalized autoregressive conditional heteroskedasticity (GARCH) structure,

$$\sigma_t^2 = \omega + \sum_{i=1}^p \alpha_i \varepsilon_{t-i}^2 + \sum_{j=1}^q \beta_j \sigma_{t-j}^2,$$

which allows volatility to depend on past squared shocks and past volatility. This formulation captures volatility clustering commonly observed in financial time series.

To allow returns to respond to observable economic and market conditions, we extend the conditional mean to include both autoregressive terms and exogenous variables. Specifically, the conditional mean is given by

$$\mu_t = \mu + \sum_{i=1}^L \phi_i r_{t-i} + \boldsymbol{\beta}^\top \mathbf{x}_{t-1},$$

where $\mathbf{x}_{t-1}$ denotes a vector of exogenous variables observed at time $t - 1$.

Combining the conditional mean and variance specifications yields an AR-X GARCH(p, q) model. By incorporating exogenous information in the conditional mean, the model allows returns to adjust directly to observable covariates while preserving a flexible, data-driven specification for time-varying volatility. Improving the specification of the conditional mean also yields residuals that are more informative for modeling conditional variance, which may enhance volatility forecasting performance.

During forecasting, both approaches produce one-step-ahead predictions $\widehat{RV}_{t+1}$, which are then evaluated using the QLIKE loss.

4 Feature Importance and Incorporation of Weather Data

A key objective of this project is to determine whether weather anomalies contain incremental predictive power for realized volatility in the energy sector. To evaluate the usefulness of potential predictors, we adopt a greedy feature-selection procedure based on out-of-sample performance measured by the QLIKE loss.

Let $\mathcal{X}$ denote the set of candidate exogenous features, which includes various measures of weather anomalies (e.g., temperature deviations, precipitation shocks, wind anomalies, and regional extreme-weather indicators), as well as other macro econometric features (fed funds, natural gas rate, VIX). Each feature is lagged appropriately so that only information available at time t is used when forecasting RV_{t+1} .

Greedy Feature Selection Procedure. We employ the following iterative scheme, using GARCH(1, 1) as the baseline specification:

1. Estimate a standard GARCH(1, 1) model using only past returns, and compute its out-of-sample QLIKE loss on the validation set. Denote this loss by L_{base} .
2. For each candidate feature $x \in \mathcal{X}$, augment the GARCH model with x_t as an exogenous regressor in the variance equation, re-estimate the model, and compute the corresponding out-of-sample QLIKE loss $L(x)$.
3. Compute the improvement in forecast accuracy

$$\Delta(x) = L_{\text{base}} - L(x),$$

and rank features according to the magnitude of $\Delta(x)$.

4. If desired, the procedure may be repeated by adding the most informative feature to the model and evaluating the marginal contribution of the remaining features.

In this framework, weather variables enter as exogenous predictors in the variance equation. By evaluating their incremental reduction in QLIKE, we can assess whether extreme temperature deviations, storm activity, or other meteorological shocks have measurable effects on energy-sector volatility. Features that consistently reduce the QLIKE loss are retained for use in the final forecasting models (including XGBoost), while features with negligible or negative contributions are excluded.

This approach enables a transparent, data-driven method for identifying which weather indicators meaningfully enhance volatility forecasts.

5 Results

Before estimating models with time-varying volatility, we first assess whether the return series exhibits conditional heteroskedasticity. If volatility is constant over time, a GARCH-type specification is unnecessary. To formally evaluate the presence of volatility clustering, we apply Engle's ARCH test to the return series.

The ARCH test evaluates whether the squared residuals from the mean equation exhibit serial correlation, which would suggest time-varying conditional variance. Under the null hypothesis of homoskedasticity, lagged squared residuals should not have explanatory power for current squared residuals. In this case, the test yields a p-value of 0.03, which is below the 0.05 significance level. Therefore, we reject the null hypothesis and find statistical evidence of conditional heteroskedasticity, supporting the use of GARCH-type volatility models.

We work with daily close-to-close returns and compute realized volatility using a 30-day (monthly) rolling window over the past five years. The resulting realized volatility series is shown below:

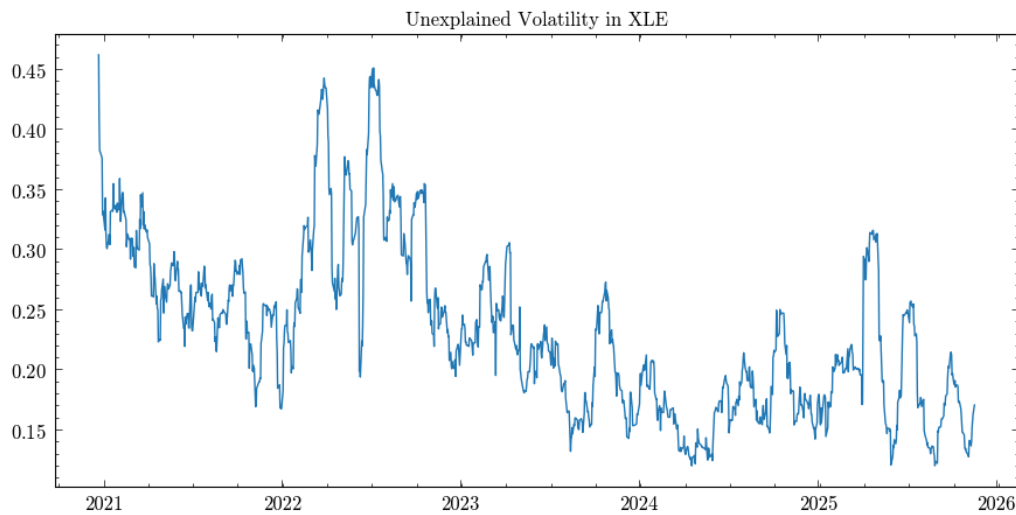


Figure 1: Monthly realized volatility constructed from daily returns.

Feature Screening via GARCH(1,1). Using the base GARCH(1,1) model as our benchmark, we conduct greedy feature selection to identify exogenous variables that reduce the out-of-sample QLIKE loss on the validation set.

Table 1: Validation-Set QLIKE for Single-Feature AR-X GARCH Models

Feature	QLIKE
Base Model (No Exogenous Variables)	1.672980
Rain	1.660200
Rain – Z-Score Deviation	1.637120
TED Spread	1.630909
Temperature	1.624912
Propane Wholesale Price	1.618738
10-Year Treasury Yield	1.613104
Retail Gasoline Price	1.611576
USD/EUR Exchange Rate	1.610767
Effective Federal Funds Rate	1.610359

From this set, we retain the top five exogenous variables to limit model complexity and mitigate overfitting. To further address potential multicollinearity, we examine pairwise correlations among the selected candidates and retain only the stronger performing variable within highly correlated pairs. This screening procedure yields the following final set of exogenous features:

$$\mathbf{x}_{t-1} = [\text{Rain}, \text{TED Spread}, \text{Temperature}, \text{Propane Wholesale Price}].$$

Model Estimation. Using the selected features, we estimate an AR-X GARCH model. Cross-validation on the validation set yields the following optimal hyperparameters:

- **Model specification:**

Volatility process : GARCH(3, 1),
Mean dynamics : AR(2) with exogenous covariates,
Innovation distribution : Skewed Student- t .

- **Validation performance:**

Best cross-validated QLIKE = 1.57.

Final Model Performance and Interpretation Evaluating the selected AR-X GARCH model on the held-out test set yields a final QLIKE score of

$$\text{QLIKE}_{\text{test}} = 2.3709,$$

indicating improved out-of-sample volatility forecasting performance relative to the baseline GARCH specification.

The final model is estimated using maximum likelihood with a skewed Student- t innovation distribution. The AR-X GARCH specification attains a log-likelihood value of -85.48 , with corresponding information criteria of $AIC = 200.95$ and $BIC = 252.52$, and is estimated on a sample of 230 observations. Table 2 reports the estimated model parameters.

Table 2: Final AR-X GARCH Model Estimates

Parameter	Estimate	Std. Error	t -stat	p -value
<i>Mean Equation</i>				
Constant	-0.0602	0.212	-0.284	0.776
r_{t-1}	0.0038	0.069	0.055	0.956
r_{t-2}	-0.1490	0.064	-2.339	0.019
Rain	-0.0994	0.047	-1.985	0.047
TED Spread	0.1289	0.052	2.492	0.013
Temperature	2.9858	1.704	1.752	0.080
Propane Wholesale Price	-0.0011	0.003	-0.380	0.704
<i>Innovation Distribution</i>				
Skewness (λ)	-0.1563	0.153	-1.024	0.306

Several covariates in the conditional mean equation exhibit statistically significant effects. The second lag of returns enters with a negative and statistically significant coefficient, providing evidence of short-term mean reversion in returns. Among the exogenous variables, the TED spread is positive and significant, indicating that periods of elevated interbank funding stress are associated with higher subsequent returns. Rainfall also enters significantly with a negative coefficient, consistent with weather-related disruptions affecting market activity. Other covariates, including temperature and propane wholesale prices, are not statistically significant at conventional levels.

Although the skewness parameter of the innovation distribution is not statistically significant, allowing for non-Gaussian innovations improves overall model fit relative to a Gaussian benchmark. Taken together, these results demonstrate that incorporating exogenous information in the conditional mean enhances volatility forecasting performance, even when exogenous variables do not enter the variance equation directly.

References

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